



RAFFLES INSTITUTION

H2 Computing (9569)

2026 Year 6

Data Protection, Ethics & Social Issues Tutorial Guide

1(a)

Backup

- Copy of current data
- Recovery of system failure

Archive

- Copy of old/unused data
- For long term storage, for future reference
- Can reduce file size once removed
- May need a program to retrieve data

1(b)

Backup procedure

- Backup complete set of current data
- At regular intervals, e.g. daily
- Stored in different media
- Store it elsewhere

Archive procedure

- When necessary
- Select data to archive (may need a program)
- Usually data is compressed to reduce file size

1(c)

- Permanent data loss
- Operation downtime

1(d)

- System performance degradation
- Non compliance to legal regulation

2(a)

- Audit trail provides traceability – can revert to previous stable version
- Collaboration in teams – without overlapping work

2(b)

- Improves readability of files, folders, variables, functions, etc
- Better organisation without confusing file names



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3. Pick the code of conduct that you can explain the most according to the context. Do not repeat the point in explaining different areas of conduct.

No.	Areas of conduct	Some questions that can be asked
1.	Integrity and Honesty	Would it violate principles of honesty and integrity? Who owns the code? Does it compromise professional judgement by potentially misleading new employer about origin of code?
2.	Responsible Use of Technology	Is it a responsible behaviour?
3.	Confidentiality and Privacy	Does it breach confidentiality agreements and compromise privacy of client or organisational data?
4.	Competence	Does it demonstrate competence in developing original solutions?
5.	Professional Conduct	Does it reflect poorly on professional conduct?
6.	Respect for Intellectual Property	Does it violate principle of respecting intellectual property rights? Copyrights, patent, trademarks, trade secrets?
7.	Public Interest and Social Responsibility	Does it respect rights and interests of former employer? Does it have negative consequences for both former employer and community?
8.	Continuing Professional Development	Does the employee update his knowledge and skills? Understand and adhere to legal and ethical standards? Seek guidance on appropriate course of action?

- (a) Property rights: breach of Intellectual Property Law
- (b) Integrity: Code belongs to previous company, which pays for the salary, hardware, development of the code
- (c) Legal action against confidentiality of code



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4(a)

- Consent: Organisations must obtain clear and informed consent before collecting, using, or disclosing personal data.
- Purpose limitation: Personal data must be collected only for specific and reasonable purposes.
- Protection: Organisations must implement security measures to prevent unauthorised access, disclosure, or loss of data.
- Retention limitation: Personal data should not be kept longer than necessary.

4(b)

- Act swiftly to contain the breach – contact the wrong recipient to request they disregard and permanently delete the email and sensitive file.

4(c)

- May face financial penalty imposed by PDPC.
- Reputational damage – loss of consumer trust.

5. In the context of maintaining user trust.

No.	Areas of conduct	Possible explanations
1.	Integrity and Honesty	Users need to have clear understanding of capabilities, limitations, and potential risks associated with autonomous vehicles. Developers must communicate openly and truthfully for users to know.
2.	Responsible Use of Technology	Design should show minimising of risk of accidents and prioritise human safety above all else. Rigorous testing done to adhere to safety standards.
3.	Public Interest and Social Responsibility	Ensure safety of users and others on the road. Reduce congestion by optimising routes. Reduce carbon footprint and environmental impact. Comply with laws and regulations, such as speed limits.



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6.

Positive Social Impacts

- Communication & Connectivity
- Entertainment & Content Access
- Health & Well-being
- Convenience & Safety

Negative Social Impacts

- Privacy Concerns
- Cyber Risks
- Misinformation
- Addiction & Mental Health

7.

Positive Impacts

- Efficiency & Automation
- Collaboration & Flexibility
- Data-Driven Decisions
- New Business Models & Jobs

Negative Impacts

- Job Displacement – taken by robots
- Skill Shifts – need to learn new skills

8.

(a) Refer to examples of rule on page 5 of notes. Each rule given should be different from one another to be awarded credit.

(b) An employee claims to be able to program in a language they have never used. Need honesty, integrity, transparency in representing one's skills and abilities.

When testing not to blame another colleague for missing an area they were supposed to test. Be professional, be constructive – communicate, collaborate, learn from mistakes, not blaming.



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9.

No.	Areas of conduct	Examples of rule
1.	Integrity and Honesty	Integrity means being truthful and honest in professional interactions. By knowingly allowing a potentially catastrophic error to go unaddressed, the manager may be acting in a manner contrary to the principle of integrity.
2.	Responsible Use of Technology	Responsibility to prioritize the safety and well-being of users and the public. To delay fixing a critical fault in the control system software could be viewed as a failure to fulfill this responsibility.
3.	Competence	Competence involves maintaining and updating professional knowledge and skills to ensure the quality and safety of one's work. By dismissing the severity of the fault and suggesting it can be addressed later, the manager may demonstrate a lack of understanding of the potential consequences and the urgency of the situation.
4.	Professional Conduct	Professionalism encompasses conducting oneself in a manner that upholds the reputation and integrity of the profession. The manager's response may be viewed as unprofessional. Professionalism requires prioritizing ethical considerations and acting with integrity, responsibility, and competence in all professional endeavors.
5.	Public Interest and Social Responsibility	Potentially catastrophic consequences could result from the manager's decision.

10.

Confidentiality and Privacy: Private data collected from client should not be leaked and be destroyed after a regulated period of time

Responsible use of technology: Software must be thoroughly tested and upgraded swiftly when necessary to ensure vehicles can operate safely

Public interest and social responsibility: Routes travelled must be safe and visible to public, stopping at designated stops that are safe for passengers and road users.